

MODULE 13

Security Monitoring & SIEM Basics

Phase 3: Security Operations Fundamentals

SOC Analyst / Tier 1 Focus

Cybersecurity Training Program

Module Overview:

This module is about **how a SOC continuously watches an organization's systems, collects security events, identifies suspicious activity, and investigates alerts using a SIEM.**

1. Learning Objectives

By the end of this module, you should be able to:

- Explain security monitoring and why a SOC needs it.
- Explain the difference between an event, log, alert, incident, and finding.
- Describe the basic role of Tier 1, Tier 2, and Tier 3 SOC analysts.
- Identify common security log sources such as Windows, Linux, firewall, DNS, web server, and EDR logs.
- Explain what a SIEM does and why organizations use one.
- Understand collection, normalization, correlation, alerting, investigation, and response.
- Explain True Positive, False Positive, False Negative, and True Negative.
- Understand basic SIEM use cases such as brute force, after-hours access, data exfiltration, and C2 beaconing.
- Follow a beginner-friendly Tier 1 alert triage process.
- Install and navigate Splunk at an introductory level and write basic SPL searches.
- Analyze log evidence and produce a short SOC-style triage summary.

2. Why This Module Matters

Earlier modules taught you how systems, networks, Linux, and logs work. In Phase 3, you start using that knowledge the way a security analyst does. Instead of only asking "What does this command do?", you now ask: "What happened, is it suspicious, and what should a SOC analyst do next?" This shift in mindset from technical knowledge to analytical judgement is what separates a beginner from a working analyst.

Career Connection: Tier 1 SOC Analyst is a common entry-level security role. Typical work includes monitoring alerts, investigating logs, gathering context, documenting findings, and escalating confirmed or complex incidents. Everything in this module builds a skill you will use in that exact job.

3. What Is Security Monitoring?

Security monitoring is the continuous process of collecting and analyzing information from computers, servers, networks, applications, cloud services, and security tools in order to identify suspicious activity.

Simple definition: Security monitoring = continuously watching security-related activity so that suspicious behavior can be detected and investigated.

Why do we need monitoring?

- Organizations generate thousands or millions of events every day.
- Humans cannot manually inspect every single event.
- Attackers try to hide their actions inside normal, everyday activity.
- Monitoring helps detect suspicious patterns earlier, before major damage happens.
- Monitoring creates evidence that analysts can investigate later.

Understand it with an example

Imagine a company with 500 computers. Every computer produces events every minute a user logs in, a firewall blocks traffic, a web server receives requests, an administrator changes a setting. Can one person manually watch everything happening across 500 machines, all day, every day? No. Security monitoring gives the SOC a systematic way to collect and analyse this activity instead of relying on one person staring at screens.

Quick Check: Why is monitoring continuous instead of something we do once a week? Because attacks can happen at any time, so security teams need ongoing visibility, not a weekly check.

4. What Is a SOC?

A Security Operations Centre (SOC) is a team, process, and technology function responsible for monitoring, detecting, analysing, and responding to cybersecurity threats.

Think of a SOC as a security control room: Sensors and systems generate data → monitoring tools collect it → alerts are produced → analysts investigate → incidents are contained and resolved.

SOC Tiers

Tier	Main Role	Typical Work
Tier 1	Alert Analyst	Monitor alerts, validate, gather basic context, classify, document, escalate.
Tier 2	Incident Responder	Deep investigation, build timeline, determine scope, containment, technical analysis.
Tier 3	Threat Hunter / SME	Proactive hunting, advanced detection, forensics, complex incidents.

What does a Tier 1 analyst ask?

- What happened?

- When did it happen?
- Which user, host, IP, or account is involved?
- Was the action successful?
- Is the activity normal or suspicious?
- What evidence supports my conclusion?
- Should I close the alert or escalate it?

Keep these seven questions in mind you will use them constantly, both in this module's labs and in real SOC work.

5. Events, Logs, Alerts, Incidents & Findings

These five words are used constantly in SOC work, and beginners often confuse them. Learn the difference carefully.

Term	Beginner Meaning	Example
Event	Something happened.	A user entered a wrong password.
Log	A record of events produced by a system.	Linux records a failed SSH login.
Log Entry	One record/line representing an event.	One line saying "Failed password".
Alert	A warning generated because a rule or condition was triggered.	10 failed logins from one IP.
Incident	A confirmed security problem that needs response.	Investigation confirms brute force and compromise.
Finding	An observation discovered and documented during investigation.	The same IP targeted five accounts.

The easiest way to remember

```

Something happens      -> EVENT
System records it      -> LOG
Detection rule notices it -> ALERT
Analyst confirms it    -> INCIDENT
Analyst records it     -> FINDING

```

Worked Example: SSH Brute Force

```

09:00:01 Failed SSH login
09:00:05 Failed SSH login
09:00:08 Failed SSH login
...
09:01:00 Many failures from the same IP
09:01:03 Successful login

```

One failed login alone is not enough to prove an attack everyone mistypes a password occasionally. A repeated pattern followed by a successful login is much more suspicious, because it suggests an attacker tried many passwords and eventually guessed correctly.

6. Security Monitoring Lifecycle

Security monitoring is not one action it is a repeating cycle of eight stages. Understanding this flow helps you see where your work as an analyst fits into the bigger picture.

Stage	What Happens	Output
1. Collect	Gather logs and telemetry.	Raw security data
2. Normalize	Put different data formats into consistent fields.	Structured/searchable data
3. Correlate	Look for relationships between events.	Suspicious pattern
4. Alert	Generate an alert when detection conditions are met.	Alert/ticket
5. Triage	Tier 1 reviews and classifies the alert.	TP/FP/escalation decision
6. Investigate	Determine timeline, scope, cause, and impact.	Investigation findings
7. Respond	Contain, eradicate, recover, and communicate.	Incident resolution
8. Improve	Tune rules and learn from the incident.	Better detection/process

Analogy — CCTV control room: Collect = cameras record. Normalize = recordings go into one common system. Correlate = several cameras show the same person. Alert = the control room receives a warning. Triage = the guard checks whether it is suspicious. Investigate = review more footage. Respond = stop the threat. Improve = adjust the cameras or rules.

7. Alert Outcomes: TP, FP, FN, TN

Outcome	Meaning	Security Example
True Positive (TP)	Alert fired and a real threat exists.	10 failed logins are confirmed as brute force.
False Positive (FP)	Alert fired but activity is benign.	User mistyped password repeatedly.
False Negative (FN)	No alert fired but a real threat existed.	Attacker logs in using stolen valid credentials and no rule detects it.
True Negative (TN)	No alert and no threat exists.	Normal employee login produces no alert.

Most dangerous: False Negative is especially dangerous a real attack exists, but the detection system never tells the SOC about it. This is why tuning detection rules carefully matters so much.

Alert Fatigue

Alert fatigue happens when analysts receive too many alerts, especially false positives. Over time, analysts may become overloaded and start treating alerts too quickly and carelessly. This increases the chance of missing a real attack hidden among the noise.

Ways to reduce alert fatigue

- Tune detection rules so they are more accurate.
- Use thresholds and context instead of alerting on every single event.

- Create allowlists for known legitimate activity.
- Prioritize alerts by severity and confidence.

8. Alert Severity

Severity	Meaning	Typical Action
INFO	Normal/audit information.	No immediate action.
LOW	Minor or informational anomaly.	Routine review.
MEDIUM	Notable suspicious activity.	Investigate within the organization's SLA.
HIGH	Serious threat.	Prompt investigation and escalation if confirmed.
CRITICAL	Likely active compromise or severe impact.	Immediate investigation/containment.

Important: Severity rules differ between organizations. Always follow the organization's SOP (Standard Operating Procedure) and escalation policy rather than assuming a fixed rule applies everywhere.

9. Log Sources Where Monitoring Data Comes From

Source	What It Can Show	Example Security Use
Linux authentication logs / journal	SSH logins, sudo/authentication activity.	Brute force or suspicious login.
Windows Security Event Log	Logons, privilege use, account changes.	Suspicious account or privilege activity.
Firewall	Source/destination IP, port, protocol, allow/deny.	Scanning or blocked attacks.
DNS	Domain lookups from endpoints.	C2 or suspicious domains.
VPN	Remote login and connection information.	Compromised credentials.
Web server	HTTP requests, URLs, status codes, user agents.	Scanning, SQL injection indicators.
EDR/Antivirus	Processes, malware detections, file activity.	Endpoint compromise.
Cloud audit logs	API and administrative actions.	Cloud account abuse.

Important for our Kali lab: On current Kali systems, traditional files such as /var/log/auth.log may not exist, depending on configuration. systemd journal data can often be queried with journalctl instead. Always check the actual logging configuration on your system rather than assuming a file exists.

```
journalctl
journalctl | grep ssh
journalctl -u ssh
journalctl --since today
```

The exact log source depends on the operating system and how it is configured — always verify before you assume.

10. How to Read a Security Log

Example Linux-style SSH log entry:

```
Apr 28 09:44:22 webserver01 sshd[1423]: Failed password for root
from 203.0.113.45 port 52341 ssh2
```

Field	Value
Time	Apr 28 09:44:22
Host	webserver01
Service	sshd
User	root
Source IP	203.0.113.45
Source Port	52341
Action	Failed password
Protocol	ssh2

Analyst habit: extract the important fields

- When?
- Who?
- From where?
- Against what?
- What action?
- Success or failure?
- How many times?
- What happened before/after?

Train yourself to answer these questions automatically every time you look at a log line. This habit is the foundation of good triage.

Windows Event IDs Beginner Set

Event ID	Meaning	Why Analysts Care
4624	Successful logon	Unexpected source/time/logon type may indicate compromise.
4625	Failed logon	Repeated failures can indicate brute force/password spraying.
4672	Special privileges assigned	Can indicate privileged account activity.

Event ID	Meaning	Why Analysts Care
4698	Scheduled task created	Can be used for persistence.
4720	User account created	Unexpected accounts may be attacker persistence.
4728	User added to security group	Could indicate privilege escalation.
7045	New service installed	Common persistence mechanism.

11. What Is a SIEM?

SIEM stands for Security Information and Event Management. It is a platform that collects security data from many sources, makes the data searchable, detects suspicious patterns, generates alerts, and helps analysts investigate.

Simple definition: SIEM = one central security monitoring platform that brings together logs and helps analysts find suspicious activity.

Why not just open every log file manually?

- There may be too many systems and too much data to check by hand.
- Different systems use different log formats.
- Correlation across multiple systems is difficult to do manually.
- Searching and filtering are much faster inside a SIEM.
- Alerts can be generated automatically instead of requiring constant watching.
- Dashboards provide a quick visual view of security activity.
- Investigations can be documented and repeated consistently.

Core SIEM capabilities

Capability	Simple Explanation
Collection	Receive data from systems and tools.
Parsing/Normalization	Turn different formats into useful, consistent fields.
Indexing/Storage	Store data for fast searching.
Correlation	Connect related events to detect patterns.
Alerting	Notify analysts when conditions are met.
Search	Let analysts investigate historical events.
Dashboards	Show important activity visually.
Threat Intelligence	Add context about known malicious indicators.

12. Correlation Rules

A correlation rule is logic that looks at one or more events and decides whether their combination is suspicious. This is what turns a pile of raw events into a meaningful alert.

Simple example

```
IF 10 failed logins
AND same source IP
AND within 60 seconds
THEN generate HIGH alert
```

A single failed login is common and usually harmless. Ten failures from the same IP in one minute is far more meaningful. Ten failures followed by a successful login is even more suspicious. Correlation gives the analyst context that individual events cannot provide on their own.

Common rule types

Rule Type	Example
Threshold	More than 10 failed logins in 60 seconds.
Sequence	Multiple failures followed by success.
Allowlist/Blacklist	Connection to a known malicious IP.
Baseline	User downloads far more data than normal.
Velocity	A sudden large increase in DNS requests.
Multi-stage	Archive creation followed by large outbound transfer.

13. Common SIEM Use Cases

Use Case A Brute Force

An attacker repeatedly attempts passwords to gain access.

```
10+ authentication failures
from same source IP
against same account
within 60 seconds
```

Triage steps: identify the source IP, target account, failure count, whether a successful login followed the failures, threat intelligence on the IP, and related firewall activity.

Use Case B Suspicious After-Hours Access

A privileged account logs in at an unusual time. This may be completely legitimate, but it still deserves context and investigation.

Check whether the user is on-call, whether the source device is expected, what was accessed, and whether other suspicious events occurred around the same time.

14. Tier 1 Alert Triage

Triage means quickly and systematically deciding what an alert means and what should happen next.

Step	Action
1. Receive	Confirm the alert and timestamp.
2. Read	Understand exactly what triggered it.
3. Gather Context	Search related events around the time.
4. Enrich	Check IP/domain/account/device context.
5. Classify	TP, FP, or escalate.
6. Respond	Follow the approved playbook.
7. Document	Record evidence and reasoning.
8. Close/Escalate	Close with justification or hand off clearly.

The 5W + H investigation habit

Question	Example
Who?	Which user/account?
What?	What action occurred?
When?	What time and duration?
Where?	Which host/source/destination?
Why?	Is there a legitimate reason?
How?	What technique or sequence was used?

Golden rule: Do not classify an alert from one line alone when more context is available. Build a small timeline around the alert before you decide.

15. Introduction to Splunk

Splunk is a platform used to collect, search, analyse, visualize, and alert on machine-generated data. In SOC work, analysts commonly use Splunk to investigate security events.

Basic Splunk architecture

Component	Simple Role
Forwarder	Collects and sends data.
Indexer	Receives, indexes, and stores data.
Search Head	Provides the interface for searches and investigation.
Deployment Server	Helps manage forwarder configuration in larger environments.

Splunk interface — what you will see

- Search bar — write SPL queries here.
- Time picker — controls the time range of your search.
- Search results — the matching events.
- Fields — extracted fields you can filter or group by.
- Dashboards — saved visual views.
- Alerts — searches that can trigger notifications.

16. Full Setup Guide: Installing Splunk on Your Kali Lab

Before the hands-on Splunk labs later in this module, set up your own local Splunk instance. This guide uses Splunk Enterprise's free trial, installed as a .deb package, which fits our Kali/Debian-based lab environment.

Note: Run these steps inside your lab VM, not on a machine you rely on for daily use. The free trial runs as a full-featured Enterprise license for the trial period and then switches to a free tier with some feature limits — this is fine for practice.

Step 1 — Check system requirements and update Kali

Splunk needs a reasonable amount of RAM (at least 2–4 GB recommended for lab use) and free disk space. Update your system first:

```
sudo apt update && sudo apt upgrade -y
```

Step 2 — Create a free Splunk account

- Go to splunk.com and click "Free Splunk" / "Get My Free Trial".
- Create an account with your email (a training/lab email is fine).
- Once logged in, open the Splunk Enterprise download page.

Step 3 — Download the Linux .deb package

On the download page, select the Linux tab and choose the .deb package option (Kali is Debian-based, so .deb is the right format). You can either click Download, or copy the wget command shown on the page to download directly inside your VM:

```
wget -O splunk-latest-linux-amd64.deb \  
"<paste the exact wget link Splunk shows you after login>"
```

The filename and version number change often, so always copy the exact link from your account's download page rather than typing an old version number from memory.

Step 4 — Install the package

From the folder where the file downloaded (often ~/Downloads):

```
cd ~/Downloads  
sudo apt install ./splunk-latest-linux-amd64.deb
```

The Splunk Enterprise .deb package always installs into /opt/splunk. This must be a regular directory, not a symbolic link.

Step 5 — Start Splunk and accept the license

```
cd /opt/splunk/bin
sudo ./splunk start --accept-license
```

The first time you run this, Splunk will ask you to create an administrator username and password. Choose credentials you will remember — you will use them every time you log in during the labs.

Step 6 — Log in to Splunk Web

- Open a browser inside your Kali VM.
- Go to <http://localhost:8000>
- Log in with the admin username and password you just created.

Tip: If port 8000 does not load, confirm Splunk actually started successfully by running: `sudo /opt/splunk/bin/splunk status`

Step 7 — Load sample security data for practice

You need some log data to search before the SPL exercises make sense. Two easy options:

- Option A — Use your own lab logs: forward SSH/auth logs from your Kali VM using a Universal Forwarder, or simply upload a saved log file (e.g. exported journalctl output) directly through Splunk Web → Add Data → Upload.
- Option B — Use a public sample dataset: Splunk's own "Boss of the SOC" (BOTS) datasets or the built-in Splunk tutorial data are good free datasets for practicing security searches — search for them from the Splunk Web "Add Data" page or the Splunkbase app directory.

When uploading a file, set the sourcetype (for example, `linux_secure` or a custom sourcetype) and index (for example, `security`) so that the example searches in this module — which use `index=security` — work correctly. If your instructor provides pre-built lab data or a shared index name, use that instead of creating your own.

Step 8 — Confirm everything works

In the Splunk Web search bar, run:

```
index=* | head 10
```

If you see events returned, your Splunk instance is collecting and indexing data correctly, and you are ready for the SPL practice and hands-on lab sections later in this module.

Common setup problems

Problem	Likely Cause / Fix
apt install fails on the .deb file	Make sure you are in the correct download folder and the filename in the command matches exactly (use tab-completion).
localhost:8000 does not load	Splunk may still be starting — wait a minute, then check: <code>sudo /opt/splunk/bin/splunk status</code>

Problem	Likely Cause / Fix
Forgot admin password	Reset it by editing <code>/opt/splunk/etc/system/local/user-seed.conf</code> before first start, or consult Splunk's official password-reset docs for your version.
No results for <code>index=security</code>	The index or sourcetype you searched does not exist yet — check Settings → Indexes, or adjust your search to <code>index=*</code> first.
Low disk space warnings	Splunk indexes can grow quickly; delete old test data or increase your VM's virtual disk size.

17. SPL: Splunk Processing Language

SPL is Splunk's search language. A pipe (`|`) sends the results of one command into the next command, letting you build a search step by step.

```
index=security "Failed password" | stats count by src_ip | sort -count
```

Read it as:

1. Search the security data for failed-password events.
2. Count the events by source IP.
3. Sort from highest count to lowest.

Beginner SPL commands

Command	Purpose	Example
<code>index=</code>	Choose the data index.	<code>index=security</code>
<code>sourcetype=</code>	Filter by data type.	<code>sourcetype=linux_secure</code>
<code>stats</code>	Count/group results.	<code>stats count by src_ip</code>
<code>sort</code>	Sort results.	<code>sort -count</code>
<code>head</code>	Keep first N results.	<code>head 10</code>
<code>table</code>	Display selected fields.	<code>table _time src_ip user</code>
<code>where</code>	Filter calculated results.	<code>where count > 5</code>
<code>dedup</code>	Remove duplicates.	<code>dedup src_ip</code>
<code>timechart</code>	Show activity over time.	<code>timechart count</code>
<code>rex</code>	Extract fields with regex.	<code>rex field=_raw ...</code>

18. Beginner SPL Practice

Use the index and field names available in your own lab — they may differ from these examples, so adjust as needed.

Query 1 — Search all available data

```
index=*
```

Question: How many events are returned? What sourcetypes are available?

Query 2 — Find failed logins

```
index=security "Failed password"
```

Query 3 — Count failed logins

```
index=security "Failed password" | stats count
```

Query 4 — Count by source IP

```
index=security "Failed password" | stats count by src_ip | sort -count
```

Query 5 — Find noisy IPs

```
index=security "Failed password" | stats count by src_ip | where count > 5 | sort -count
```

Query 6 — Show useful fields

```
index=security "Failed password" | table _time src_ip user
```

Query 7 — Authentication over time

```
index=security ("Failed password" OR "Accepted password") | timechart count span=1h
```

If your lab uses different indexes, sourcetypes, or field names, adjust each query to match your dataset.

19. Hands-On Lab — Manual Alert Triage

Work as a Tier 1 SOC analyst. Read the evidence carefully first. Do not jump directly to a conclusion.

Alert #001 — Multiple Failed SSH Logins

```
Apr 28 02:17:01 web01 sshd: Failed password for root from 185.220.101.47
Apr 28 02:17:03 web01 sshd: Failed password for admin from 185.220.101.47
Apr 28 02:17:05 web01 sshd: Failed password for ubuntu from 185.220.101.47
Apr 28 02:17:07 web01 sshd: Failed password for pi from 185.220.101.47
Apr 28 02:17:08 web01 sshd: Failed password for test from 185.220.101.47
Apr 28 02:17:09 web01 sshd: Failed password for user from 185.220.101.47
Apr 28 02:17:10 web01 sshd: Failed password for guest from 185.220.101.47
Apr 28 02:17:11 web01 sshd: Failed password for oracle from 185.220.101.47
Apr 28 02:17:12 web01 sshd: Failed password for postgres from 185.220.101.47
Apr 28 02:17:59 web01 sshd: Failed password for mysql from 185.220.101.47
Apr 28 02:18:00 web01 sshd: Failed password for ftpuser from 185.220.101.47
Apr 28 02:19:44 web01 sshd: Accepted password for deploy from 185.220.101.47
Apr 28 02:19:45 web01 sshd: session opened for user deploy
```

Student Tasks

4. Classify: TP, FP, or escalate.
5. List timestamps, usernames, source IP, and successful-login evidence.
6. Decide the severity and explain why.
7. Identify the attack type.
8. Give three immediate actions.
9. Write a 3-sentence Tier 2 handoff.

Analysis: The pattern targets many common usernames rather than one user — this strongly suggests automated password guessing/password spraying-style activity. The later successful login for deploy significantly raises concern, because the same source IP moved from repeated failures to successful access.

Alert #002 — After-Hours Privileged Access

```
23:54 CORPDC01 EventID=4624 Account=j.malik LogonType=3 Source=10.0.50.22
23:54 CORPDC01 EventID=4672 Account=j.malik
      Privileges=SeBackupPrivilege,SeRestorePrivilege
23:56 FILESERV EventID=4663 Account=j.malik
      Object=Finance/Q1_Reports Process=robocopy.exe
23:57 FILESERV EventID=4663 Account=j.malik
      Object=Finance/Salaries_2026.xlsx Process=robocopy.exe
23:57 FILESERV EventID=4663 Account=j.malik
      Object=HR/Personnel_Records Process=robocopy.exe
```

Student Tasks

10. Explain Event ID 4672.
11. Explain what robocopy.exe does.
12. Identify what sensitive data was accessed.
13. Decide whether this should be closed, investigated, or escalated.
14. List additional information needed to determine legitimacy.

15. Choose an escalation severity and justify it.

Important: Do not automatically call this malicious. Suspicious activity requires investigation and context — a legitimate administrator may have had an emergency task. The evidence determines the next action, not assumptions.

20. Hands-On Lab Introductory Splunk Investigation

Suggested duration: 60–90 minutes. Use the Splunk instance you set up in Section 16, a provided lab environment, or another authorized lab.

Exercise 1 — Navigation

16. Open Search & Reporting.
17. Set the time range to "Last 24 hours".
18. Run `index=*`.
19. Record the event count.
20. Record available sourcetypes.
21. Expand one event and record five fields.
22. Check the host field and note which systems are sending data.

Exercise 2 — Failed Login Investigation

```
index=security "Failed password"

index=security "Failed password" | stats count by src_ip | sort -count

index=security "Failed password" | stats count by src_ip | where count > 5 | sort -count

index=security "Failed password" | stats count by src_ip, user | sort -count | head 10
```

Exercise 3 — Investigate One IP

Replace X.X.X.X with the source IP you identified in Exercise 2.

```
index=* src_ip="X.X.X.X"

index=* src_ip="X.X.X.X" | table _time src_ip user action
```

Then search for successful authentication using the field names available in your dataset, or search raw events if fields are not extracted:

```
index=* src_ip="X.X.X.X" ("Accepted password" OR "successful" OR "4624")
```

Exercise 4 — Timechart

```
index=security ("Failed password" OR "Accepted password") | timechart count span=1h
```

Save the search as a report if your lab supports saved reports. Add it to a dashboard and create a second panel for the top source IPs:

```
index=security | stats count by src_ip | sort -count | head 10
```

21. SOC Dashboard Basics

A dashboard is a collection of visual panels that helps analysts understand security activity quickly, at a glance.

Panel	Question It Helps Answer
Alert volume	Are alerts suddenly increasing?
Top source IPs	Which sources generate the most activity?
Authentication failures	Are login failures increasing?
New accounts	Are accounts being created unexpectedly?
Privileged activity	Are admin accounts behaving normally?
Open incidents	What requires attention now?

A dashboard does not replace investigation. It helps an analyst notice patterns and prioritize attention — you still have to dig into the details yourself.

22. Knowledge Check

23. Explain the difference between an event, log, alert, and incident using an SSH example.
24. What is alert fatigue and why is it dangerous?
25. Define TP, FP, FN, and TN. Which is especially dangerous for security monitoring?
26. What is a correlation rule? Give one example.
27. Why does a SOC use a SIEM instead of manually reading every log file?
28. What is the difference between Tier 1 and Tier 2?
29. What does the pipe character (|) do in SPL?
30. What does Event ID 4625 represent?
31. Why can a successful login after many failures be highly suspicious?
32. Give three possible reasons why a Splunk query returns zero results.

23. Student Reflection

Write at least 150 words answering: How does a SIEM change the way you think about security monitoring compared with manually reading individual log files? Which concept in this module was most challenging? Explain what you understood, what you found difficult, and what you would practice next.

24. Common Beginner Mistakes to Watch For

- Treating every failed login as an attack.
- Calling an alert an incident before investigation.
- Ignoring timestamps and surrounding events.

- Looking only at the source IP and forgetting the target account/host.
- Using a very broad time range during investigation.
- Memorizing SPL commands without understanding the data.
- Assuming a tool name or unusual User-Agent automatically proves compromise.
- Forgetting to document evidence and reasoning.
- Declaring suspicious activity malicious without checking legitimate context.

25. One-Page SOC Cheat Sheet

```

SECURITY MONITORING
Watch -> Collect -> Analyse -> Detect -> Investigate -> Respond

CORE TERMS
Event      = something happened
Log        = record of what happened
Alert      = warning generated by detection logic
Incident   = confirmed security problem
Finding    = documented observation

SIEM
Collect -> Normalize -> Correlate -> Alert -> Search -> Investigate -> Report

TRIAGE
1. Read the alert           6. Check success/failure
2. Identify source         7. Gather surrounding events
3. Identify target         8. Enrich indicators
4. Check time              9. Classify TP / FP / Escalate
5. Check volume            10. Document

SPL
index=...
sourcetype=...
| stats count by ...
| sort -count
| head 10
| table ...
| where ...
| timechart count span=1h

WINDOWS IDs TO REMEMBER
4624 = Successful logon      4720 = New user
4625 = Failed logon         4728 = Added to security group
4672 = Special privileges   7045 = New service
4698 = Scheduled task

INVESTIGATOR QUESTIONS
Who? What? When? Where? Why? How?

MOST IMPORTANT HABIT
Do not investigate only the alert.
Investigate the STORY around the alert.

```